

TESTs - Lesson 12 - Tokenized Votes

What is the primary difference between the ERC20Votes extension and a traditional ballot system with a chairperson?



- A** ERC20Votes uses token balances to determine voting power, unlike traditional ballots where a single chairperson grants voting rights.
- B** ERC20Votes requires voters to have NFTs, whereas traditional ballots use a simple token.
- C** ERC20Votes can only be deployed on specific blockchains, while traditional ballots are universal.
- D** ERC20Votes allows only the chairperson to vote, while traditional ballots allow multiple voters.

According to the ERC20Votes documentation, how does the system keep track of an account's vote power history?



- A** By maintaining a history of checkpoints related to each account's vote power.
- B** By continuously updating the vote power on every single transaction.
- C** By using NFTs to represent individual voting rights.
- D** By requiring all users to register their vote power annually.

In the ERC20Votes extension, is merely holding tokens enough to exercise voting power?



- A** The tokens were not properly minted.
- B** No, users must explicitly delegate their voting power, even to themselves.
- C** Voting power is only assigned after a certain time delay.
- D** The getVotes() function is currently bugged.

What is the main benefit of using the ERC20Permit extension for approvals?



- A** It enables off-chain approvals using signatures, avoiding gas fees for the approval step.
- B** It allows on-chain approvals without needing any gas fees.
- C** It restricts who can approve transactions, enhancing security.
- D** It simplifies token transfers by eliminating the need for approvals.

After minting 10 tokens to acc1, why does acc1 initially show 0 voting power when checked using getVotes()?

- ☐ A The tokens were not properly minted.
- ☒ B Voting power needs to be explicitly activated through self-delegation.
- ☐ C Voting power is only assigned after a certain time delay.
- ☐ D The `getVotes()` function is currently bugged.

What is the core problem that ERC20Votes aims to solve regarding using standard ERC20 tokens for voting?

- A** The lack of unique identifiers for each vote cast.
- B** The high gas fees associated with ERC20 token transfers.
- C** The susceptibility to vote buying and manipulation due to tokens being transferable before voting closes.
- D** The inability of ERC20 tokens to be transferred.

How does the ERC20Votes extension ensure that historical voting decisions remain valid, even if token balances change over time?

- A** By allowing voters to change their past votes.
- B** By keeping a history of checkpoints that record vote power at specific block numbers or timestamps.
- C** By making all tokens non-transferable after a vote.
- D** By relying on external oracle services for historical data.

What is the primary purpose of delegating voting power in the ERC20Votes system?

- A** To burn tokens once a vote is cast.
- B** To explicitly activate a user's token balance as voting power, typically for a specific period or block.
- C** To reduce the total supply of tokens.
- D** To allow a single entity to control all votes.

Which OpenZeppelin Contracts extension allows for off-chain approvals using cryptographic signatures?



A AccessControl

B ERC20Votes

C ERC20Permit

D ERC20Burnable

What is one of the benefits of ERC20Votes not automatically updating voting power on every token transfer?



A It prevents tokens from being transferred to other accounts.

B It makes token transfers cheaper by avoiding constant gas fees for vote power calculation.

C It allows for anonymous voting.

D It simplifies the overall token contract logic.